



Phase 2:

Competencies / Overarching Learning

Goals of a Theorist

Version 0.0

Goal: Achieve consensus on the overarching competencies required to perform the core theoretical tasks and responsibilities identified in Phase 1

Output: Consensus list of core competencies / overarching learning goals for theoretical psychology

Introductory note (v0.0)

Phase 1 identified eight core tasks and responsibilities of theoretical psychologists. Phase 2 asks what theorists must be able to do competently in order to perform those responsibilities.

I arrived at the competencies below by reasoning through and imagining what knowledge, skills, and capabilities/capacities are required by the Phase 1 responsibilities, and then attempting to integrate these requirements into broader theoretical competencies. Here, I characterize a competency as the ability to combine relevant knowledge, skills, judgment, capabilities, and resources in order to perform a distinct substantive theoretical function/responsibility to an adequate standard.

The list is intended to identify overarching learning goals rather than every relevant piece of knowledge, individual skill, or general reasoning capacity. Some requirements therefore appear across several competencies, while others are conditional on the kind of theoretical work being performed. Also, you do not have to agree with my reasoning behind a competency to agree with the relevance/value of the competency by itself. Of course, the reverse also applies.

Finally, **please note** that I do not (and we should not) see it as a requirement that any one theorist should fully exhibit each and every competency on this list. Nor should it be a requirement that a theorist can apply these competencies individually, without the support of any technical staff or partner theorists.



Theoretical competencies / overarching learning goals

1. **Identify and formulate explananda**

The theorist can identify, formulate, categorize, and order phenomena, capacities, or problems as candidate explananda. The theorist can distinguish established explananda from provisional patterns, speculative targets, and raw data patterns. The theorist can make their epistemic status explicit.

2. **Develop and evaluate construct architectures**

The theorist can define, differentiate, coordinate, and, where necessary, prune constructs. The theorist can build and evaluate taxonomies and ontologies. The theorist can specify what theoretical terms refer to and how constructs relate to one another.

3. **Coordinate theory and measurement**

The theorist can specify how theoretical constructs could, in principle, be operationalized, indexed, measured, or otherwise empirically evidenced. The theorist can evaluate which indicators or indices count as valid. The theorist can make explicit the assumptions connecting empirical observations or measurements to theoretical entities.

4. **Develop explanatory mechanisms and processes**

The theorist can propose, progressively specify, and refine mechanistic, functional, or process-based explanations. The theorist can distinguish preliminary sketches from more specified accounts. The theorist can relate explanations across relevant levels.

5. **Construct and represent theories and models**

The theorist can construct coherent theories, models, and, where useful, model families that make core propositions and assumptions explicit. The theorist can select and develop appropriate verbal, diagrammatic, formal, mathematical, computational, or hybrid representations. The theorist can ensure that these representations preserve the intended theoretical commitments and are sufficiently explicit and usable to support comparison and testing.

6. **Diagnose, develop, and repair theories**

The theorist can diagnose tensions, gaps, ambiguities, inconsistencies, and other limitations in existing theories. The theorist can generate and examine possible revisions or variants. The theorist can extend, constrain, deepen, or repair theories when doing so improves relevant theoretical properties such as coherence, precision, parsimony, explanatory value, scope, or testability.

7. **Derive theoretical implications**

The theorist can determine what follows from a theory or model, including how it



accounts for established phenomena and, where applicable, its qualitative or quantitative predictions, intervention predictions, counterfactual expectations, and expected failures, while making explicit the assumptions required for these derivations.

8. Appraise, delimit, and compare theories

The theorist can appraise how and how well a theory explains its intended explananda. The theorist can identify its scope, boundary conditions, and expected limits. The theorist can compare rival theories on common explanatory targets. The theorist can specify, given relevant auxiliary assumptions, what evidence would count for, against, or discriminate between them.

9. Integrate evidence and update theories

The theorist can integrate fragmented, heterogeneous, and potentially conflicting empirical, formal, conceptual, and measurement-related evidence. The theorist can determine what that evidence implies for a theory and its components. The theorist can make principled decisions about what should be retained, revised, constrained, forked, merged, or discarded.

10. Design theory-guided research programmes

The theorist can translate unresolved theoretical problems, uncertainties, derived implications, and contrasts between theoretical alternatives into a prioritized and coherently sequenced programme of theoretical and empirical inquiry. The theorist can select appropriate forms of investigation for the questions involved. The theorist can organize them so that earlier work constrains, informs, or enables subsequent theory development and testing.